SCTR'S PUNE INSTITUTE OF COMPUTER TECHNOLOGY

TECHFIESTA'26

INTERNATIONAL HACKATHON

CODE. CREATE. CONQUER

- **Problem Statement ID :** HD004
- **Problem Statement Title:** Remote Patient Monitoring For Chronic Disease Management
- **PS Domain:** Healthcare
- **Team ID:** 99
- **Team Name :** God'sPlan

VitalSync: The AI-Powered Vernacular Guardian

- **Proposed Solution**

- A **low-cost** (<₹4k), screenless wearable that monitors critical vitals (SpO2, HR, Temp) and uses **Edge AI** to detect cardiac anomalies and falls. It connects to a Multilingual Mobile App designed for rural users, breaking literacy barriers with voice-guided alerts in local dialects.

- **How it Solves the Problem:**

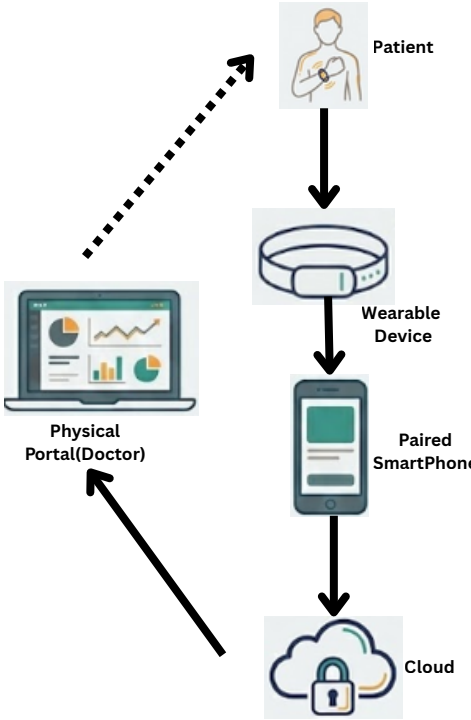
- Solves Language Barriers: Multilingual Voice UI ensures illiterate or **semi-literate** rural patients can understand medical alerts without reading.
- Solves Cost: Replaces ₹40k+ smartbands with a sub-₹4k **focused** medical device.
- Solves False Alarms: Sensor Fusion confirms falls before calling for help.

- **Detailed Breakdown**

- Wearable Node: ESP32-C3 wristband with "**Touch-to-Measure**" Copper Tape ECG and continuous PPG.
- Edge Intelligence: **TinyML algorithms** filter noise and detect Arrhythmia/Falls locally.
- Inclusive App Interface: A "**Pictorial & Voice-First**" dashboard that speaks alerts (e.g., *"Please rest, heart rate high"*) in the patient's native regional language.

- **Key Innovations**

- "**Cuffless" BP Estimation**: Novel PTT algorithm to estimate blood pressure trends.
- Vernacular Voice Engine: Dynamic Text-to-Speech (TTS) for health coaching in 10+ regional languages.
- Copper Tape Electrodes: **Low-cost fabrication** for medical-grade signals



USER FLOW DIAGRAM

TECHNICAL APPROACH

1: Signal Acquisition & Real-Time DSP

- **Precision Sampling:** Utilizes a hardware timer interrupt to sample ECG data at a strict 500Hz frequency via a custom copper-tape interface, ensuring **clinical-grade** accuracy for HRV analysis.
- **On-Chip Filtering:** Implements a **two-stage** DSP pipeline (50Hz Notch + 20Hz Butterworth) to eliminate power-line hum and muscle noise artifacts before the signal reaches the AI engine

2: Edge AI & Sensor Fusion Pipeline

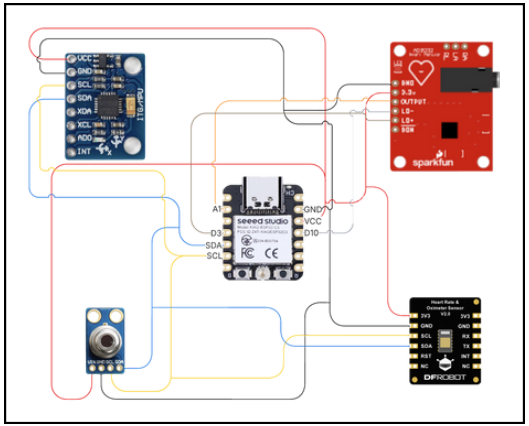
- **TinyML Inference**: Deploys a quantized **1D-CNN** model (Edge Impulse) directly on the ESP32-C3 (latency <20ms) to detect Arrhythmia patterns locally without internet dependency.
- **Multi-Modal Logic**: Uses Sensor Fusion to confirm "Fall Events" by cross-referencing high-G impact (Accelerometer) with **sudden** Heart Rate spikes, while simultaneously calculating Cuffless BP via Pulse Transit Time (PTT).

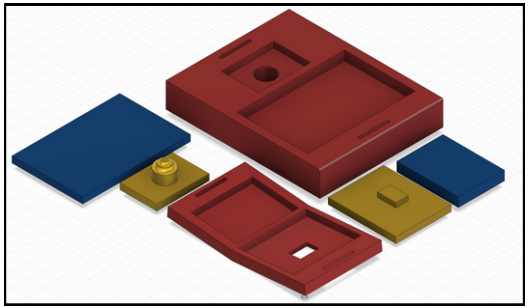
3: Resilient IoT Architecture (Store-and-Forward)

- **Offline Resilience:** buffers data to SPIFFS (Flash Memory) during **network outages** and automatically "flushes" historical logs to the MQTT broker upon reconnection (QoS 1).
- **Power Optimization:** Operates in Deep Sleep (<10μA), waking only for **scheduled** vitals checks or critical interrupt triggers (like a fall), extending battery life for rural usage.

4: Cloud Analytics & Vernacular UX

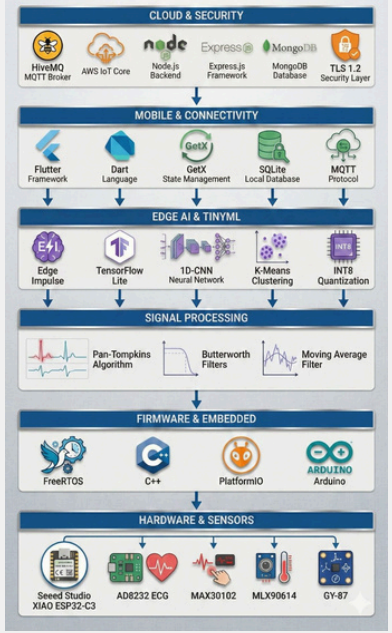
- **Longitudinal Reports**: Cloud backend aggregates data to generate "Pre vs. Post Prescription" efficacy scores, helping doctors quantify patient improvement over weeks.
- **Offline Voice Engine**: The mobile app pre-caches audio assets to deliver Vernacular Voice Alerts (e.g., in Marathi/Hindi) based on error codes, ensuring accessibility even without active data.

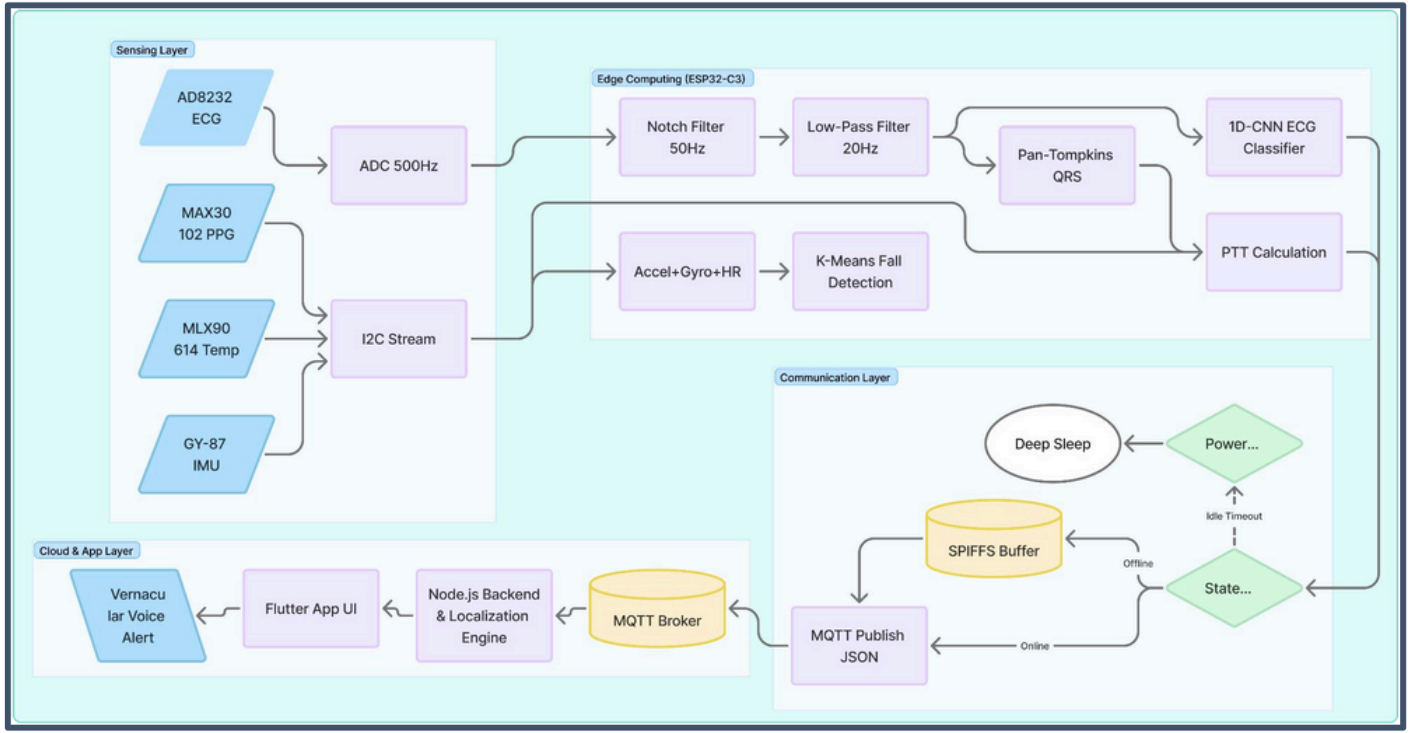




WORKFLOW

Tech Stack

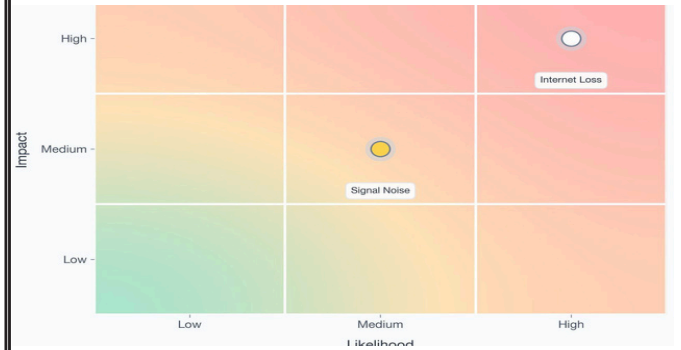




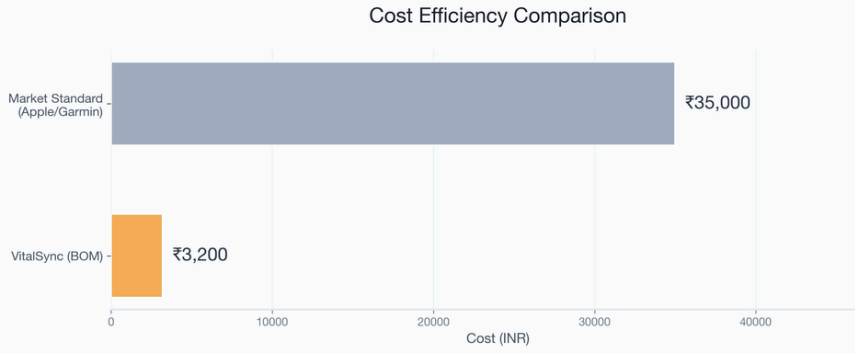
FEASIBILITY AND VIABILITY

Device Feasibility, Commercial Potential, and Scalability

Characteristic	Feasibility	Viability
Technical Maturity	ESP32-C3, standard protocols, accessible parts	OTA updates for future features
Cost Efficiency	Bill of Materials under ₹4k	Recurring revenue model
Commercial Potential	B2B healthcare, industrial pivot, consumer extensions	Strong hospital adoption probability
Scalability	MQTT architecture, edge compression	Government alignment
National Alignment	Supports ABDM, Make in India	Long-term viability boosted by OTA
New Sections	Medical-first design, modular hardware, low power, ruggedized, encryption	API access for research & insurance



Risk Analysis Metrics



IMPACT BENEFITS AND RESEARCH REFERENCE

- A Real-Time QRS Detection Algorithm (IEEE Xplore): https://ieeexplore.ieee.org/document/4122029

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- Data Sheets For Every Hardware Component:
- https://robu.in/wpcontent/uploads/2020/02/MLX90614-Datasheet-Melexis.pdf
- https://robu.in/wp-content/uploads/2014/12/GY-87_EN.pdf

- FOR MORE DETAILED PROJECT INFO: https://docs.google.com/document/d/1YGkTGM0MOvq_SjBBu33wUUh17DJuYXT7GMWttseolQ4/edit?usp=sharing